



From Slab to Beam

Tributary Areas, Line Loads and a Two-Dimensional Subframe

Course	Computer (& AI) Methods for Structural Engineering (DISC 5203)
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Assigned plan	Floor Plan 1 (final digit 1)
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24 beam segments | net slab 180.128 m² | four OpenSees models | exact area and reaction equilibrium

Scope

This report extends the existing HW1 Git working tree. It reuses the HW1 slab and vertical-element results, transfers the same gravity load to beams with 45-degree tributary boundaries, models the two marked beams as standalone continuous beams and one-floor subframes, and cross-checks subframe column forces against HW1 axial estimates.

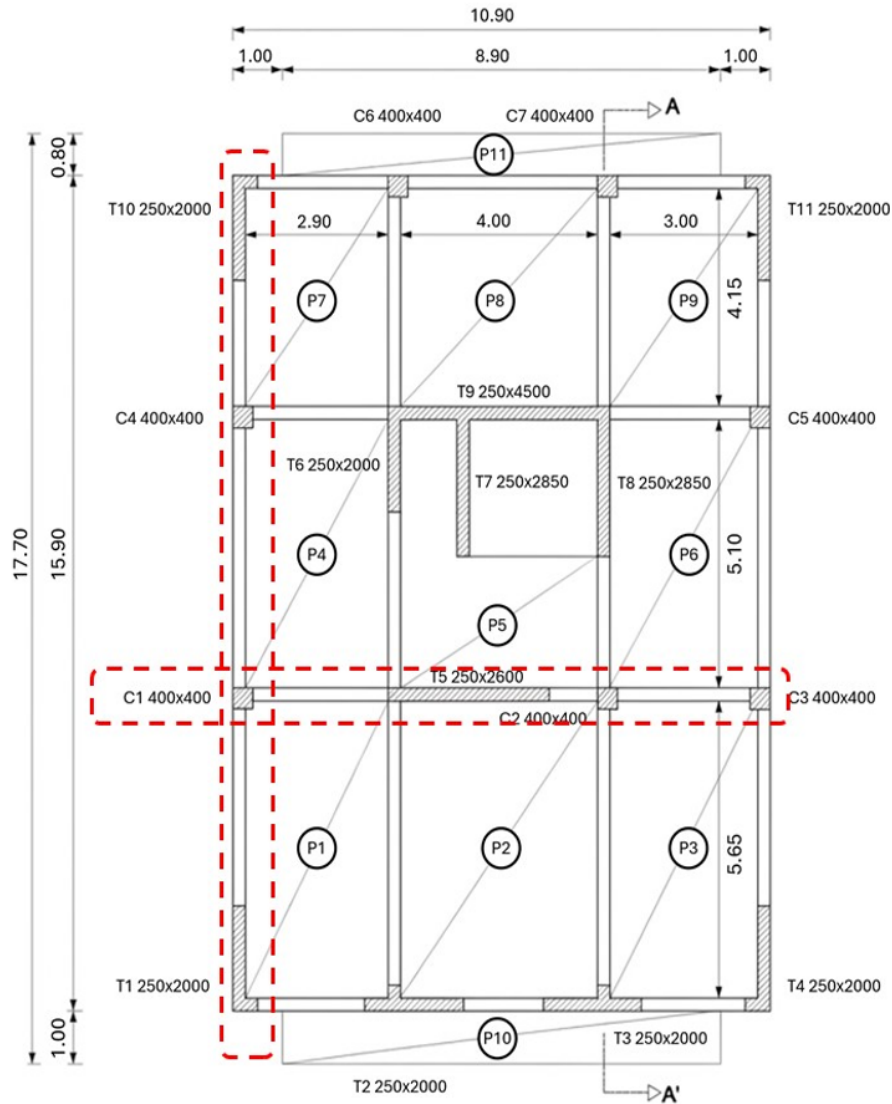


Figure 1. Assigned Floor Plan 1; red boxes mark Beam A and Beam B.

Dead slab load = $25 \times 0.150 = 3.75 \text{ kN/m}^2$; live load = 5.00 kN/m^2 . No finish load is printed. Beam self-weight is excluded from the requested slab-to-beam transfer. Unprinted beam size is explicitly assumed as $300 \times 600 \text{ mm}$; columns are $400 \times 400 \text{ mm}$, storey height 3.0 m and $E = 30.0\text{e}6 \text{ kN/m}^2$.

2. 45-degree beam tributary construction

Each cell of the unchanged HW1 net slab is assigned to its nearest finite beam edge. Parallel beams divide at mid-lines; equal distance to perpendicular edges gives $|\Delta x| = |\Delta y|$ and a 45-degree corner line. Openings are removed before assignment. P10 and P11 are cantilevers and add a uniform component to their single support line.

For a two-way rectangle with short side a and long side b : $A_{\text{short}} = a^2/4$ and $A_{\text{long}} = a(2b-a)/4$. The stored non-uniform profile, not the tabulated UDL, is used for local response.

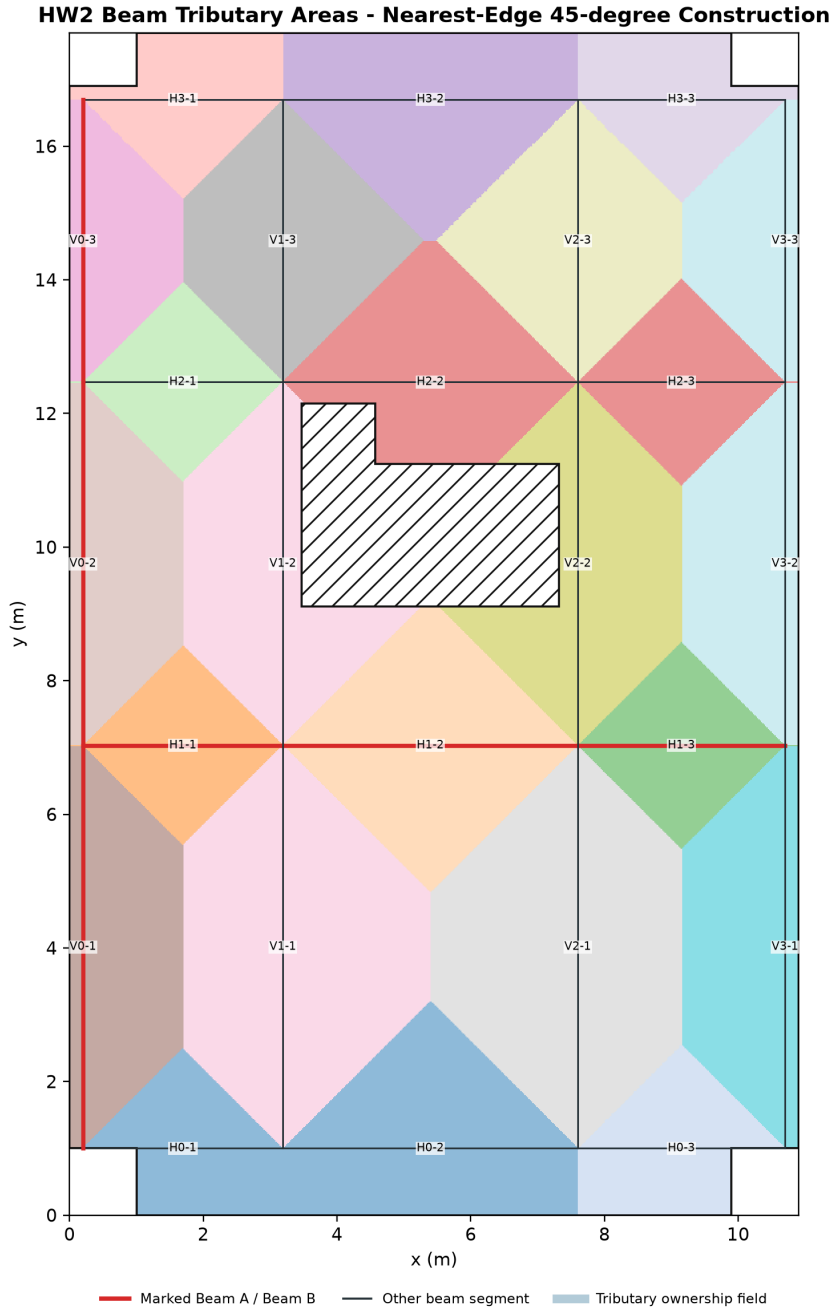


Figure 2. Beam ownership field and 45-degree divisions.

3. Beam census and equivalent line-load table

Equivalent UDL means equal total load only. It does not preserve local shear or the exact moment-diagram shape.

ID	Span	Panels	Class	Area	Dead	Live	Total
H0-1	2.990	P1; P10 (cantilever balcony where overlapping)	two loaded sides locally (main slab + balcony)	4.449	5.580	7.439	13.019
H0-2	4.410	P2; P10 (cantilever balcony where overlapping)	two loaded sides locally (main slab + balcony)	9.281	7.892	10.523	18.416
H0-3	3.100	P3; P10 (cantilever balcony where overlapping)	two loaded sides locally (main slab + balcony)	4.706	5.693	7.591	13.284
H1-1	2.990	P1; P4	interior, slab on both sides	4.473	5.610	7.480	13.089
H1-2	4.410	P2; P5	interior, slab on both sides	9.707	8.254	11.005	19.259
H1-3	3.100	P3; P6	interior, slab on both sides	4.808	5.816	7.755	13.571
H2-1	2.990	P4; P7	interior, slab on both sides	4.473	5.610	7.480	13.089
H2-2	4.410	P5; P8	interior, slab on both sides	8.210	6.981	9.308	16.289
H2-3	3.100	P6; P9	interior, slab on both sides	4.808	5.816	7.755	13.571
H3-1	2.990	P7; P11 (cantilever balcony where overlapping)	two loaded sides locally (main slab + balcony)	4.625	5.800	7.734	13.534
H3-2	4.410	P8; P11 (cantilever balcony where overlapping)	two loaded sides locally (main slab + balcony)	9.247	7.864	10.485	18.348
H3-3	3.100	P9; P11 (cantilever balcony where overlapping)	two loaded sides locally (main slab + balcony)	4.897	5.924	7.899	13.822
V0-1	6.030	P1	perimeter, slab on one side	7.982	4.964	6.618	11.582
V0-2	5.440	P4	perimeter, slab on one side	6.992	4.820	6.427	11.246
V0-3	4.230	P7	perimeter, slab on one side	4.947	4.385	5.847	10.232
V1-1	6.030	P1; P2	interior, slab on both sides	15.240	9.478	12.637	22.114
V1-2	5.440	P4; P5	interior, slab on both sides	8.980	6.190	8.253	14.443
V1-3	4.230	P7; P8	interior, slab on both sides	8.518	7.551	10.068	17.619
V2-1	6.030	P2; P3	interior, slab on both sides	15.383	9.567	12.756	22.323
V2-2	5.440	P5; P6	interior, slab on both sides	9.584	6.607	8.809	15.416
V2-3	4.230	P8; P9	interior, slab on both sides	8.652	7.670	10.227	17.896
V3-1	6.030	P3	perimeter, slab on one side	8.101	5.038	6.717	11.755
V3-2	5.440	P6	perimeter, slab on one side	7.088	4.886	6.515	11.401
V3-3	4.230	P9	perimeter, slab on one side	4.979	4.414	5.885	10.299

Units: span in m; area in m²; line loads in kN/m.

4. Load profiles and worked example

For H1-2, $L = 4.410$ m and the actual two-sided tributary area is read from P2 and P5 after the core opening is removed. $D = 3.75$ A, $L_{live} = 5.00$ A, and each equivalent line load is the corresponding total divided by 4.410 m. The figure retains the triangular/trapezoidal distribution used in OpenSees.

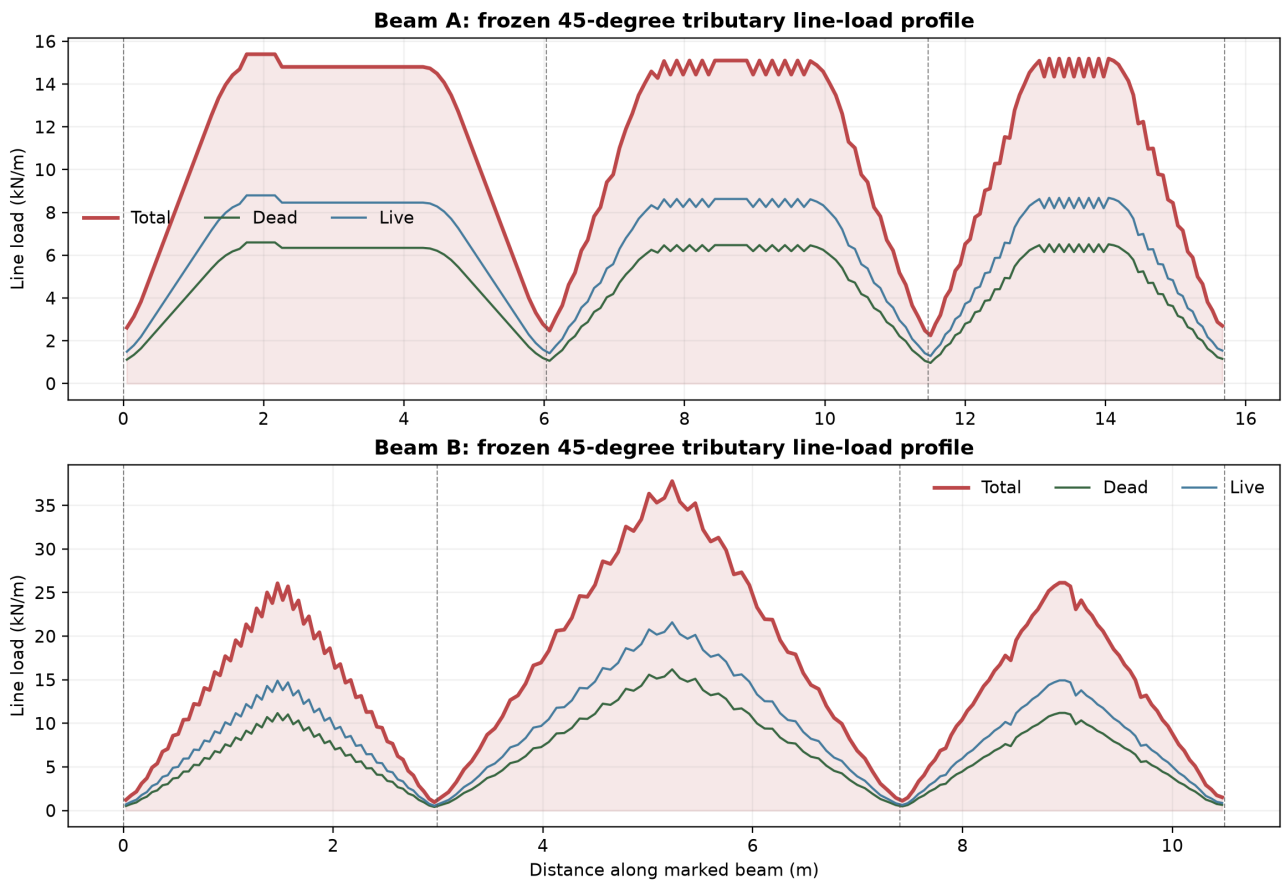


Figure 3. Frozen dead, live and total line-load profiles.

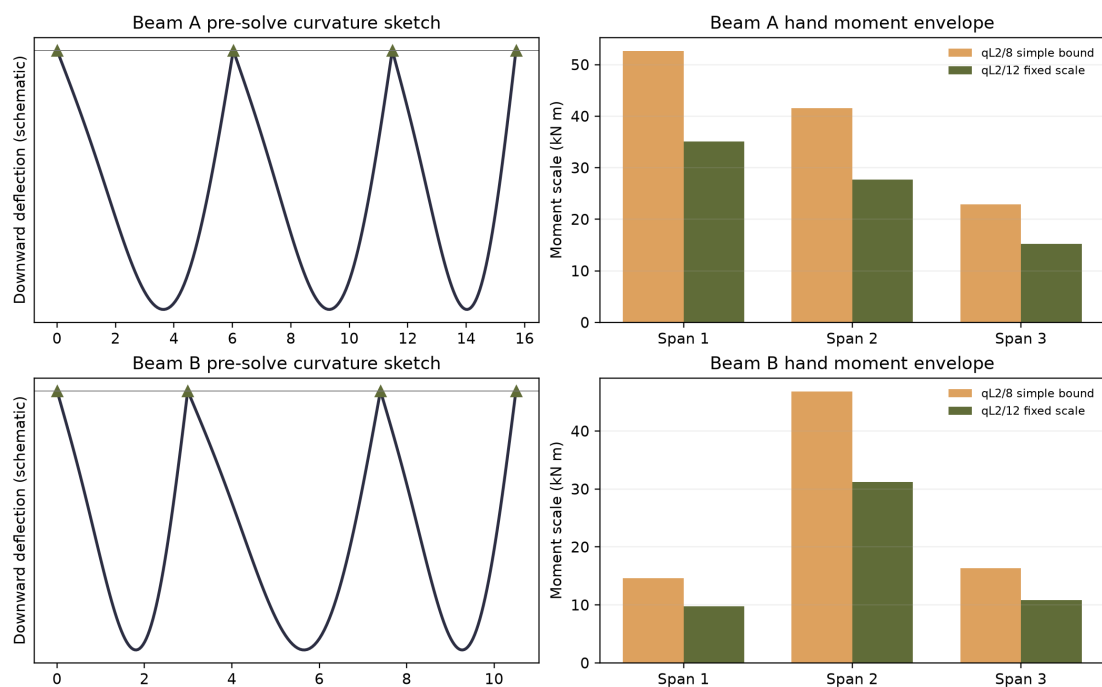


Figure 4. Pre-solve sketches and qL2 bounds.

5. OpenSees response diagrams

Standalone models retain vertical supports with free rotations. Subframes replace them with fixed-base 3 m columns sharing the beam nodes. The same frozen profile is applied in every comparison.

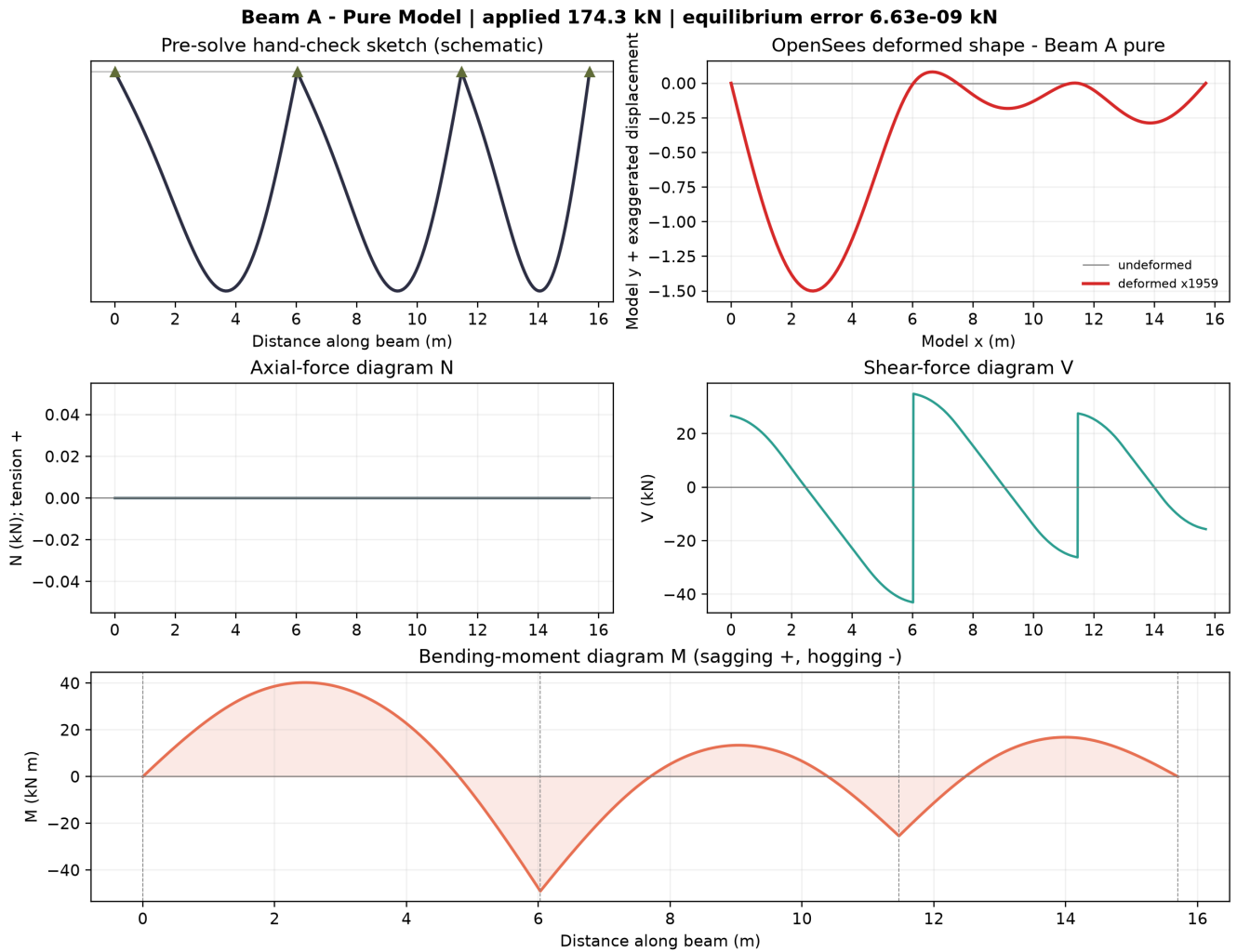


Figure. Beam A, pure model: pre-solve sketch, deformed shape and N/V/M diagrams.

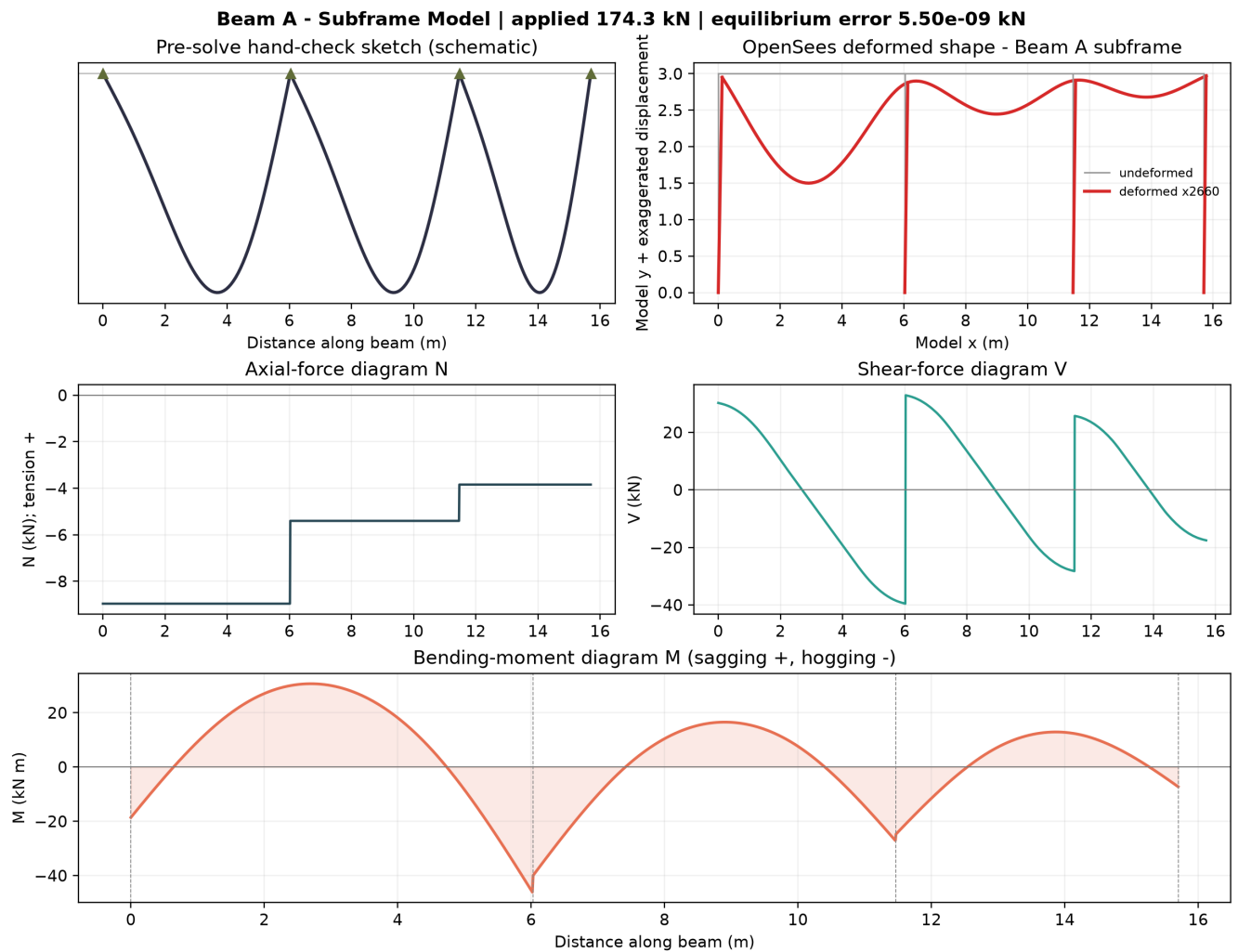


Figure. Beam A, subframe model: pre-solve sketch, deformed shape and N/V/M diagrams.

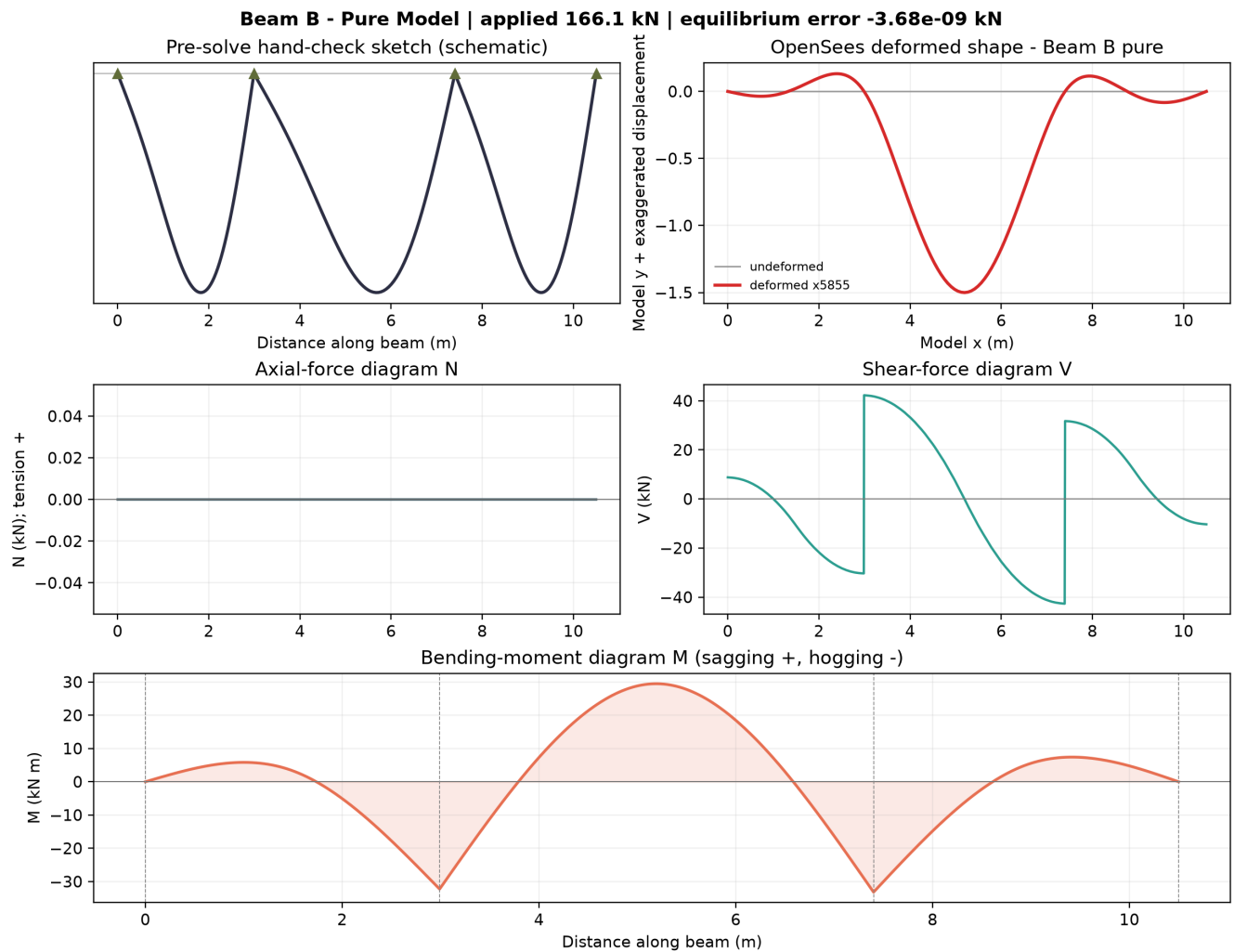


Figure. Beam B, pure model: pre-solve sketch, deformed shape and N/V/M diagrams.

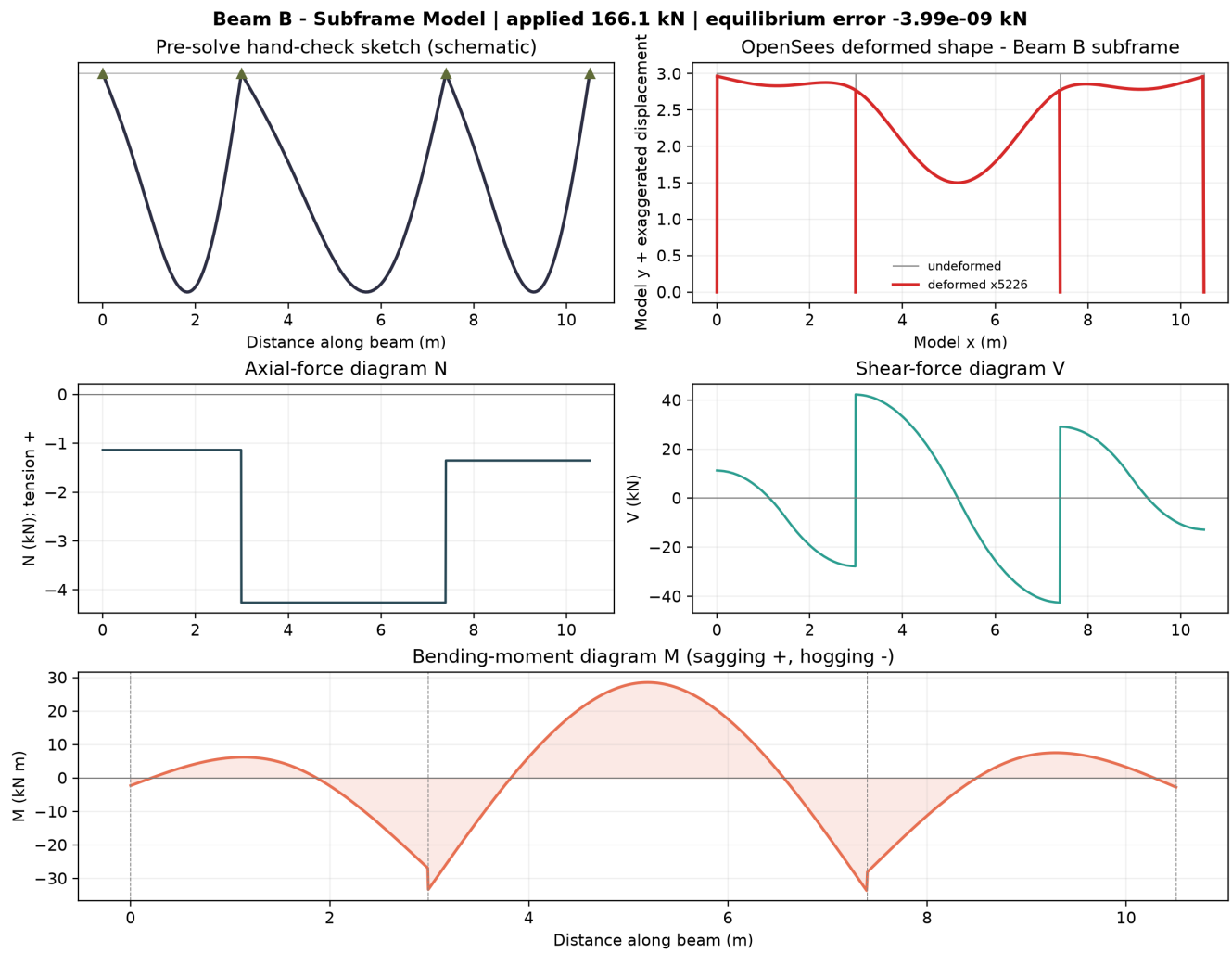


Figure. Beam B, subframe model: pre-solve sketch, deformed shape and N/V/M diagrams.

6. Numerical comparison

Beam	Model	Load	Sag +	Hog -	Max M	Down mm	qL2/8	Check
A	pure	174.305	40.243	-49.039	49.039	0.7656	52.643	PASS
A	subframe	174.305	30.678	-46.096	46.096	0.5640	52.643	PASS
B	pure	166.139	29.470	-33.191	33.191	0.2562	46.819	PASS
B	subframe	166.139	28.566	-33.604	33.604	0.2870	46.819	PASS

Loads are kN; moments are kN m. Every model closes equilibrium below $1e-8$ kN and has sagging-positive and hogging-negative regions. The subframe is the defensible beam-design model because it retains finite column stiffness; the standalone beam remains an independent bound.

7. HW1 axial cross-check

Strip	Grid	HW1 member	HW1 N	HW2 N	Diff %	Sign
A	G00	T1	-111.266	-30.239	-72.8	PASS
A	G01	C1	-32.699	-72.509	121.7	PASS
A	G02	C4	-42.360	-53.999	27.5	PASS
A	G03	T10	-66.258	-17.559	-73.5	PASS
B	G01	C1	-32.699	-11.302	-65.4	PASS
B	G11	T5	-233.992	-70.151	-70.0	PASS
B	G21	C2	-36.226	-71.814	98.2	PASS
B	G31	C3	-27.321	-12.872	-52.9	PASS

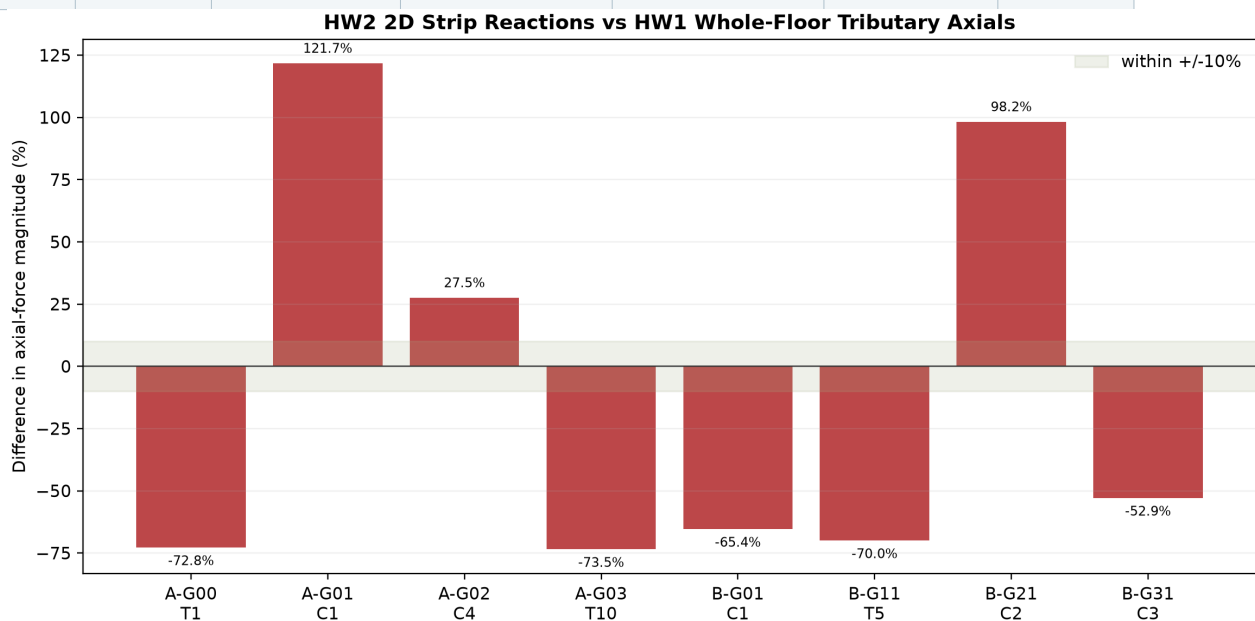


Figure. Magnitude difference between the 2D marked strips and HW1 whole-floor estimates.

Compression sign agrees at every mapped support, but the magnitude is not within a few percent. HW1 includes the whole-floor nearest-support load and aggregated walls; each HW2 result contains only one marked strip and omits perpendicular beam paths. No load was tuned. The subframe is used for the beam design question; exact cross-model reconciliation requires a 3D grid or common nodal aggregation.

8. Agent workflow and reproducibility

The word briefings were frozen before code generation. The agent transcribed them into four OpenSees entry files plus an audited shared implementation. Gates were: identify the deck, freeze loads, check units and signs, verify qL2 bounds, replace supports by columns, close equilibrium, compare HW1 axials, then leave Git commit/upload pending student review.

Run: analyze_hw2_beam_loads.py; run_hw2_models.py; summarize_hw2_models.py; build_hw2_report.py;
render_hw2_report_pdf.py; verify_hw2_outputs.py.

Limitations

The plan is raster; beam size and uncracked stiffness are assumptions; the subframes are 2D strips; and the student must add a literal handwritten/signature sheet if required. These limits are preserved rather than concealed.

Conclusion

All required beam census, tributary areas, line loads, four OpenSees cases, N/V/M and deformation diagrams, reactions, hand bounds, prompt record, HW1 cross-check and reproducibility files are present. The repository remains uncommitted and unuploaded until student sign-off.